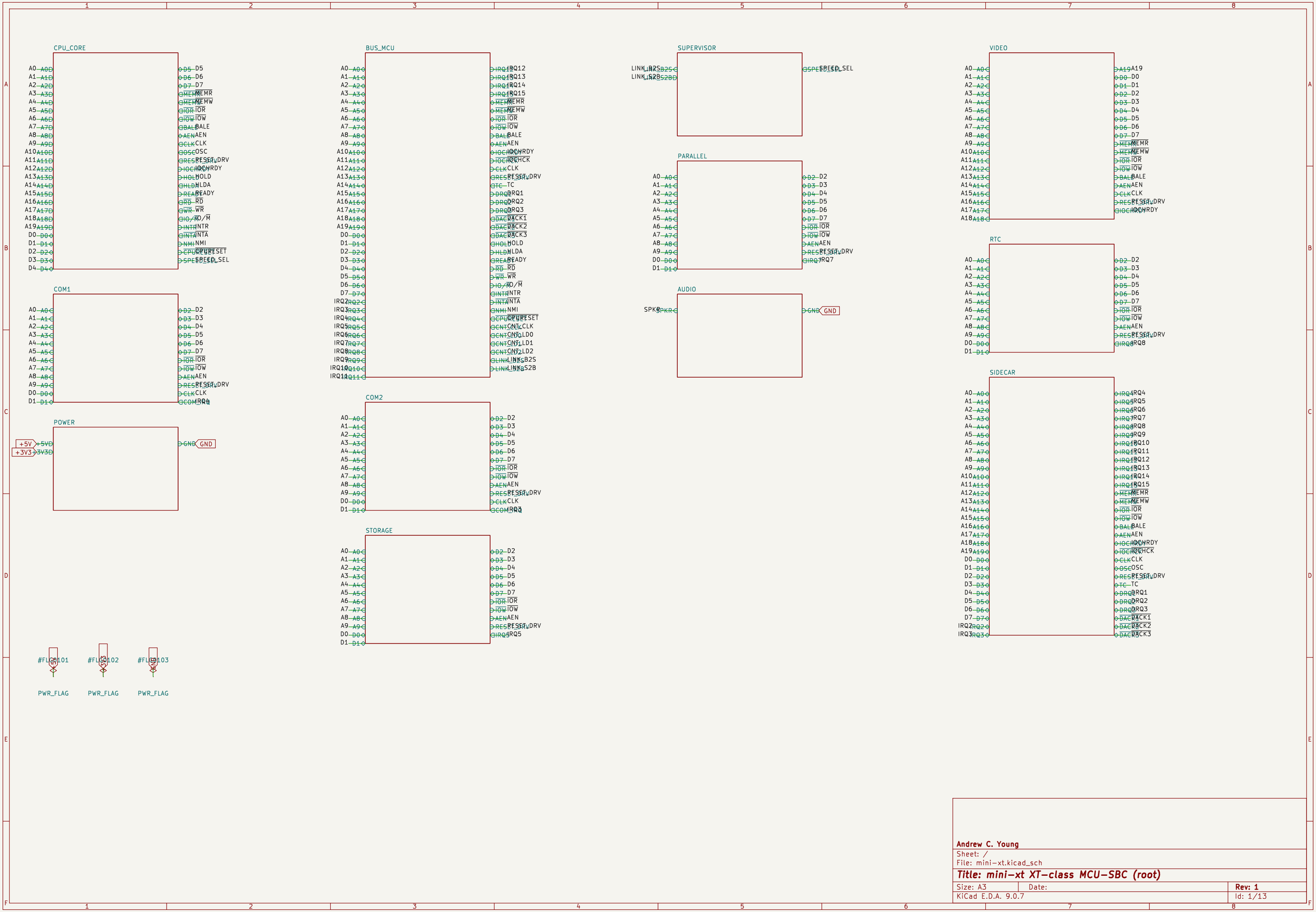


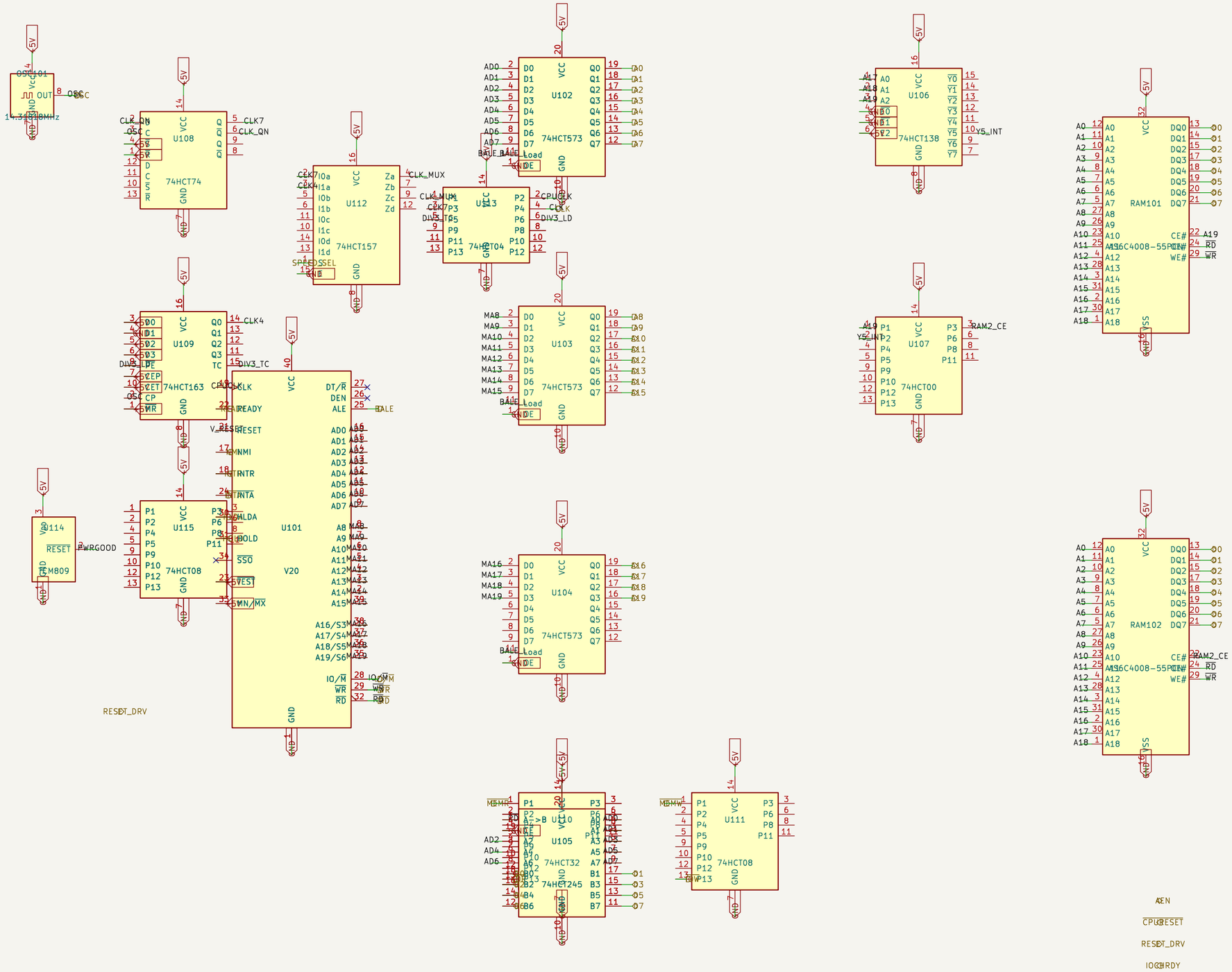
Andrew C. Young

Sheet: /

File: mini-xt.kicad_sch

Title: mini-xt XT-class MCU-SBC (root)

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 1/13



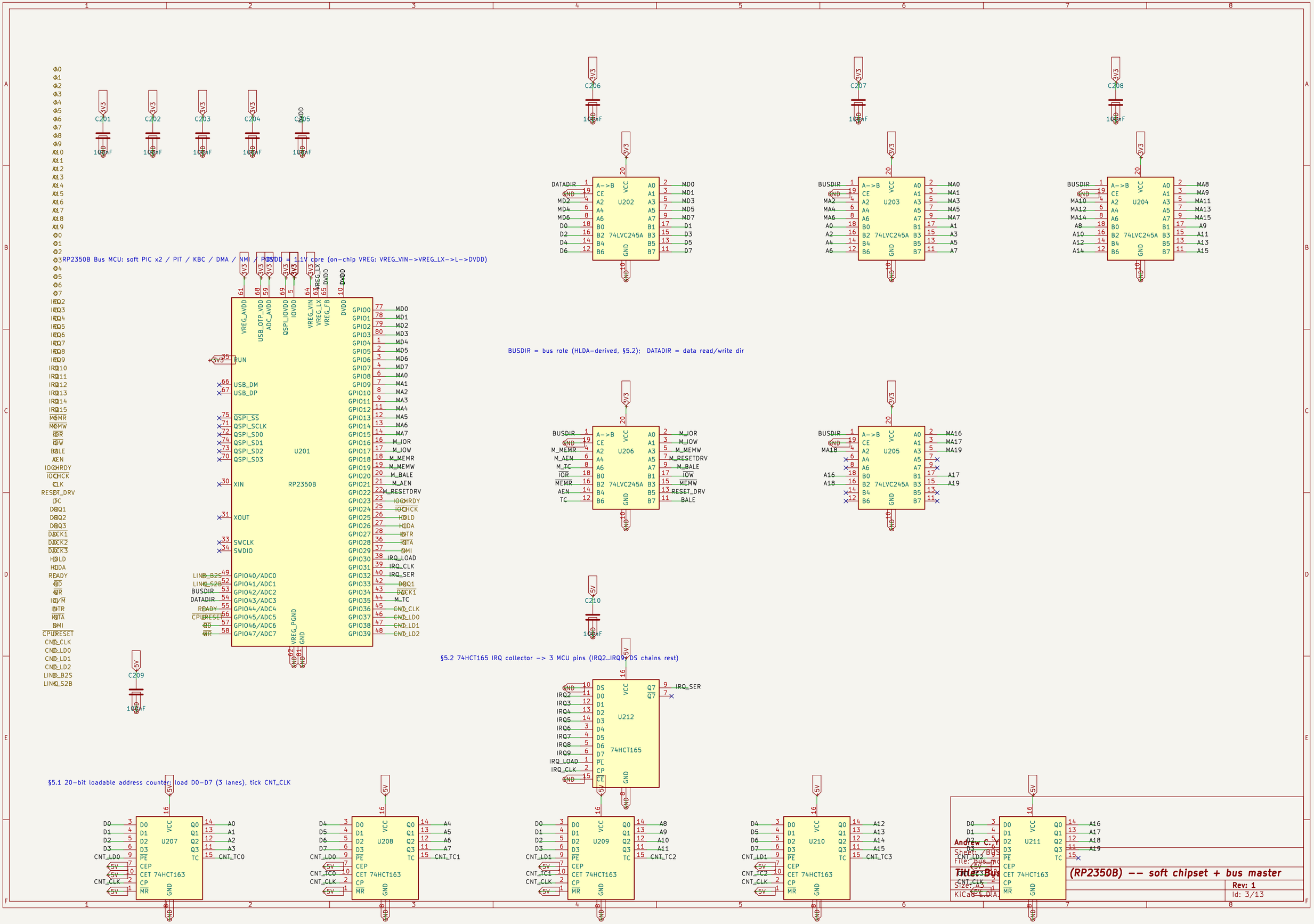
Andrew C. Young

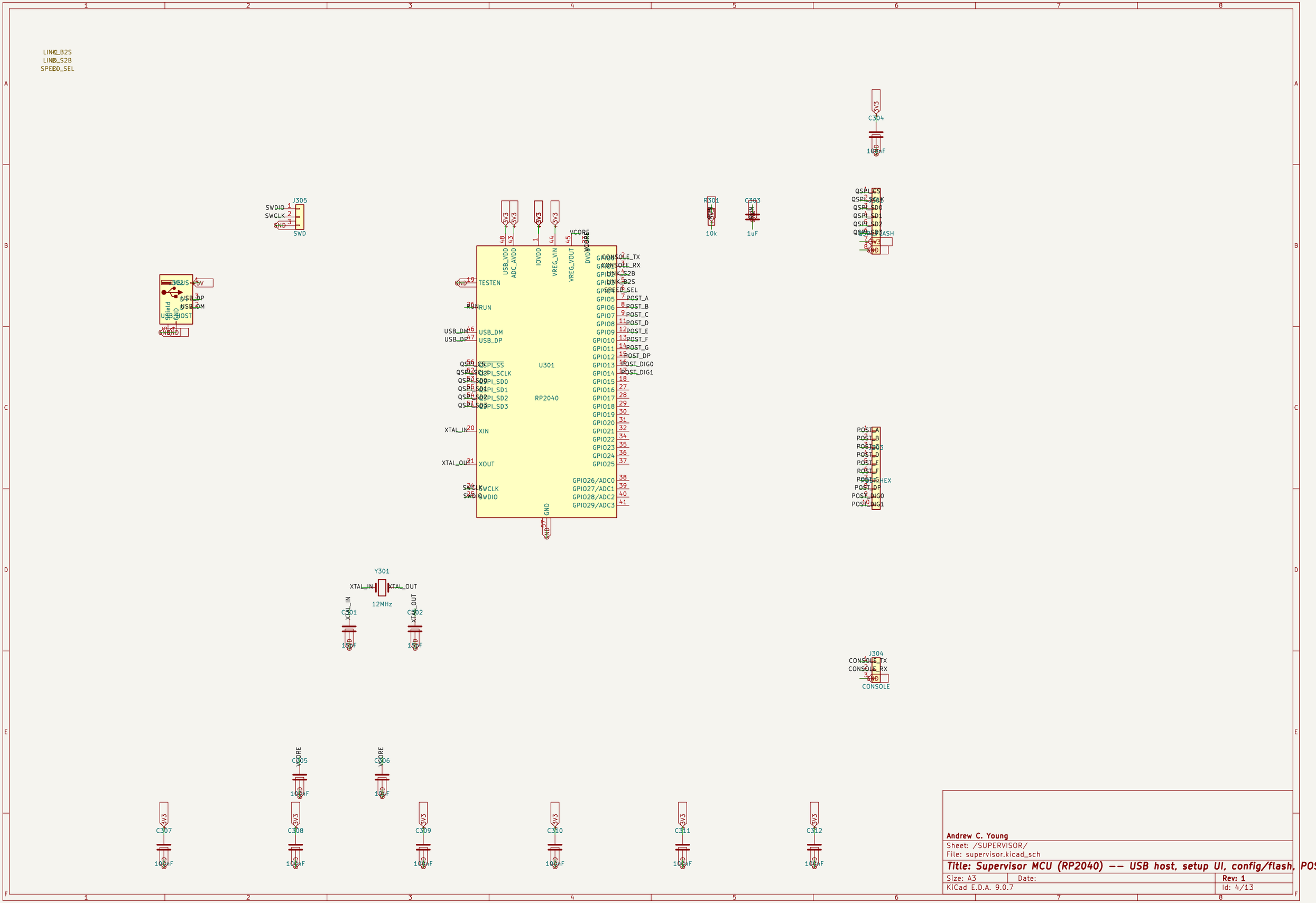
Sheet: /CPU_CORE/

File: cpu_core.kicad_sch

Title: CPU / Memory / Clock / Reset core (5V)

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 2/13


RP2350B Bus MCU: soft PIC x2 / PIT / KBC / DMA / NMI / POWDD = 1.1V core (on-chip VREG: VREG_VIN->VREG_LX->L->DVDD)
BUSDIR = bus role (HLDA-derived, §5.2); DATADIR = data read/write dir
§5.1 20-bit loadable address counter, load D0-D7 (3 lanes), tick CNT_CLK
§5.2 74HCT165 IRQ collector -> 3 MCU pins (IRQ2..IRQ9 -> DS chains rest)
U201 RP2350B
U202 74LVC245A
U203 74LVC245A
U204 74LVC245A
U205 74LVC245A
U206 74LVC245A
U207 74HCT163
U208 74HCT163
U209 74HCT163
U210 74HCT163
U211 74HCT163
U212 74HCT165
Andrew C.Y.
Sheet: /Bus
File: Bus.kicad_sch
Title: Bus
Size: A3
KiCad E.D.A.
(RP2350B) -- soft chipset + bus master
Rev: 1
Id: 3/13



Andrew C. Young

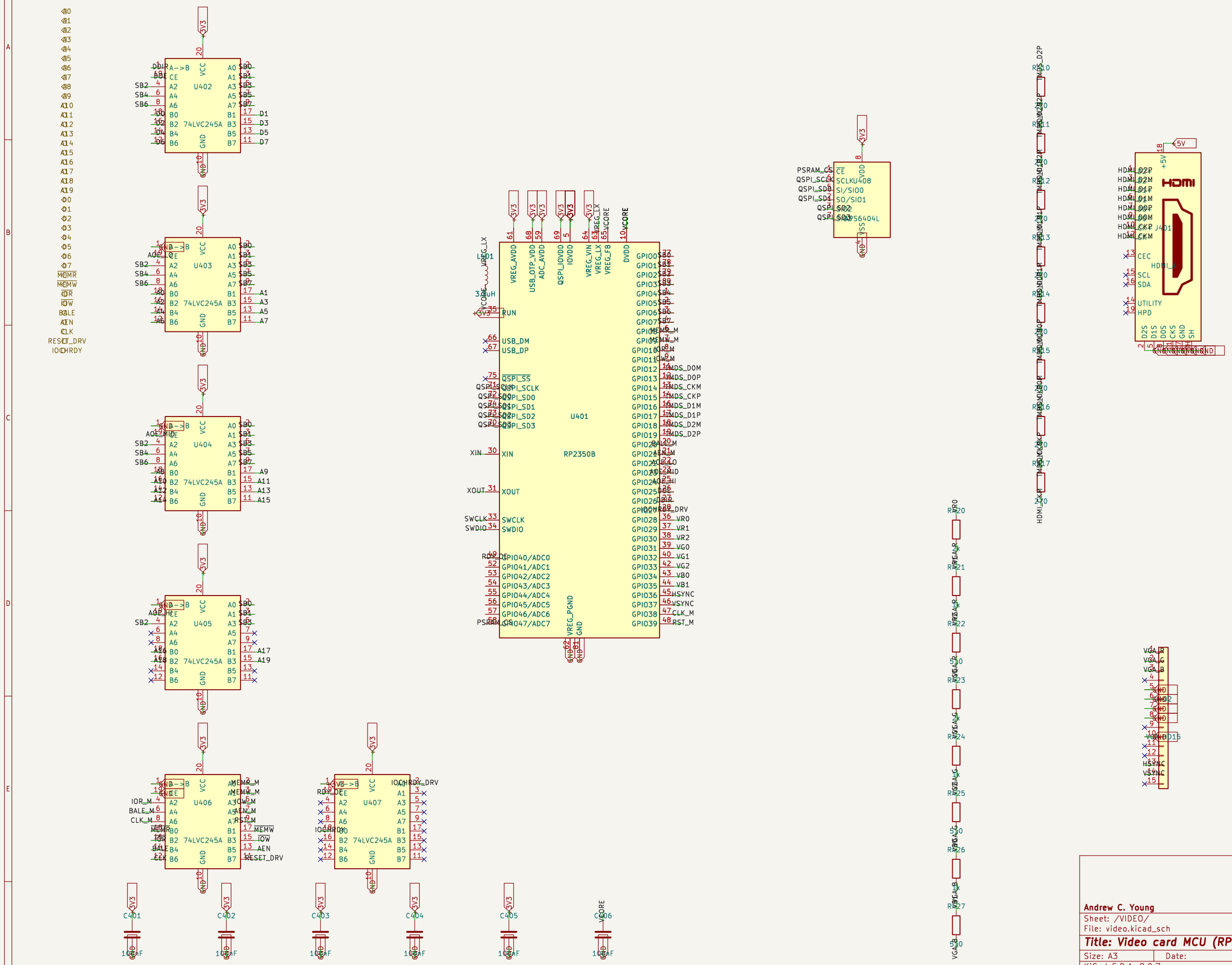
Sheet: /SUPERVISOR/

File: supervisor.kicad_sch

Title: Supervisor MCU (RP2040) -- USB host, setup UI, config/flash, POST,

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 4/13

0xBFFFF and 3B4/3B5/3B8/3BA/3BF + 3D4/3D5/3D8/3D9/3DA from snooped A17–A19/A0–A9. No Y5, no link, no host RAM (design S8).



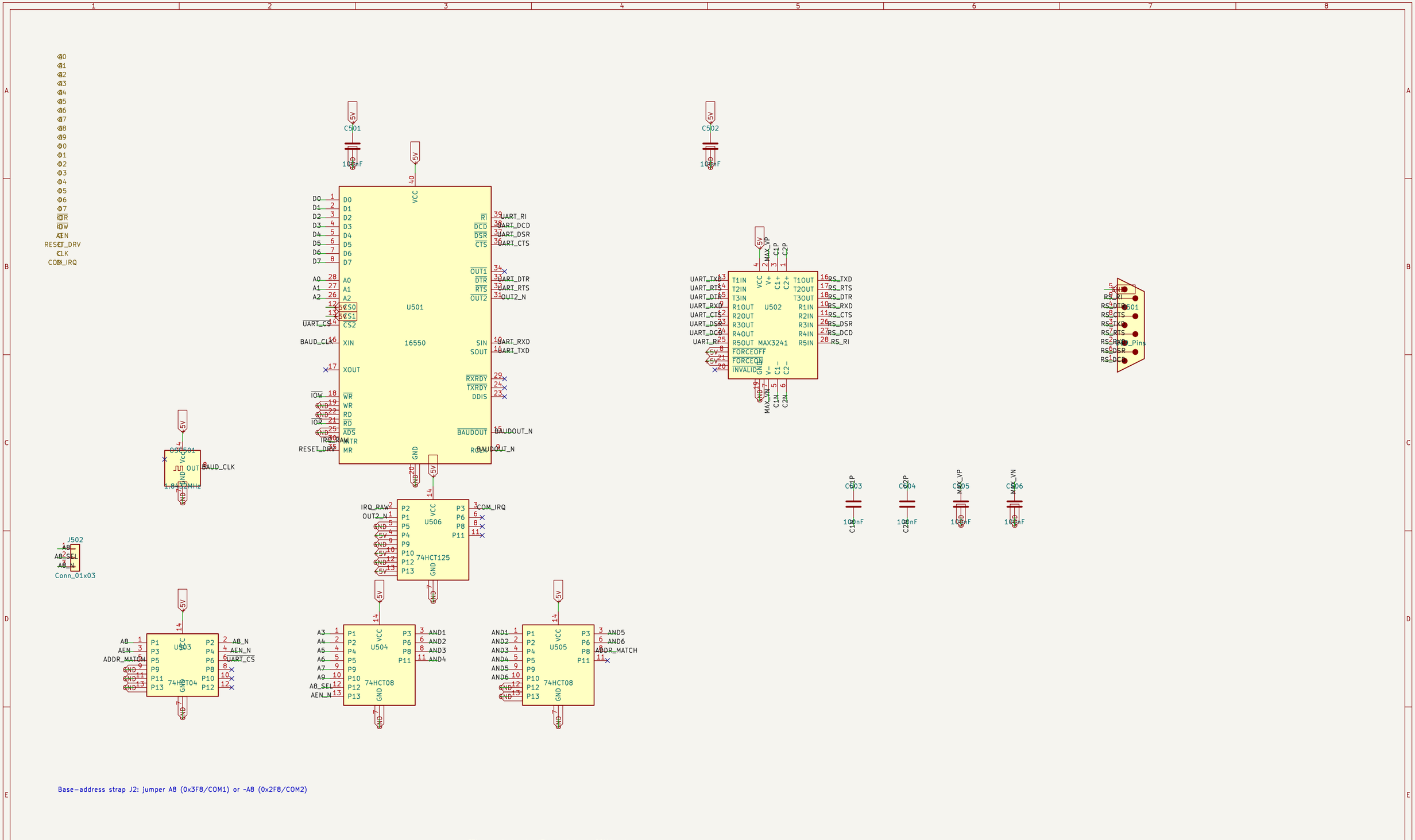
Andrew C. Young

Sheet: /VIDEO/

File: video.kicad_sch

Title: Video card MCU (RP2350B) -- soft CGA/MDA/Herc, VGA + HDMI out

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 5/13



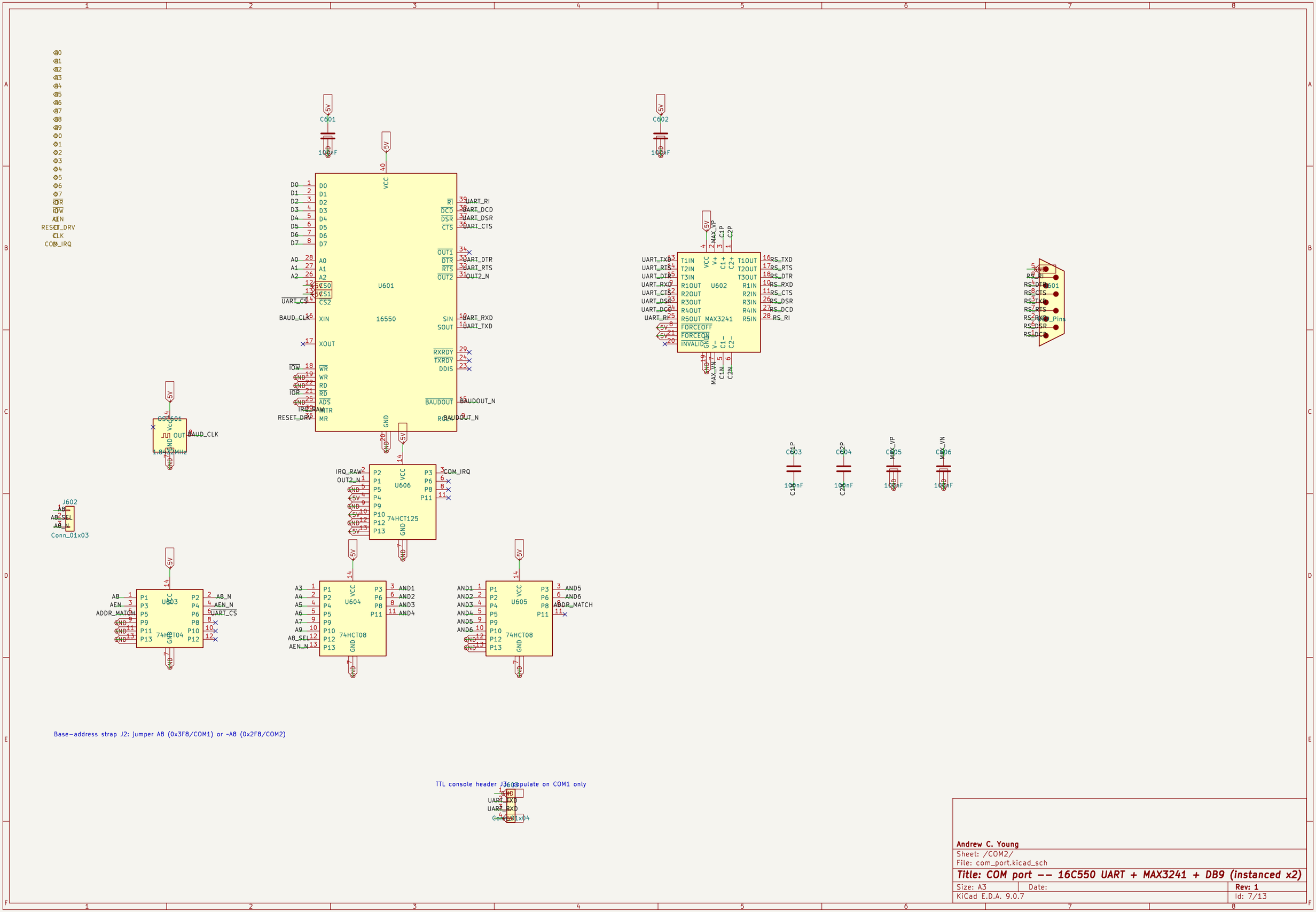
Andrew C. Young

Sheet: /COM1/

File: com_port.kicad_sch

Title: COM port -- 16C550 UART + MAX3241 + DB9 (instanced x2)

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 6/13



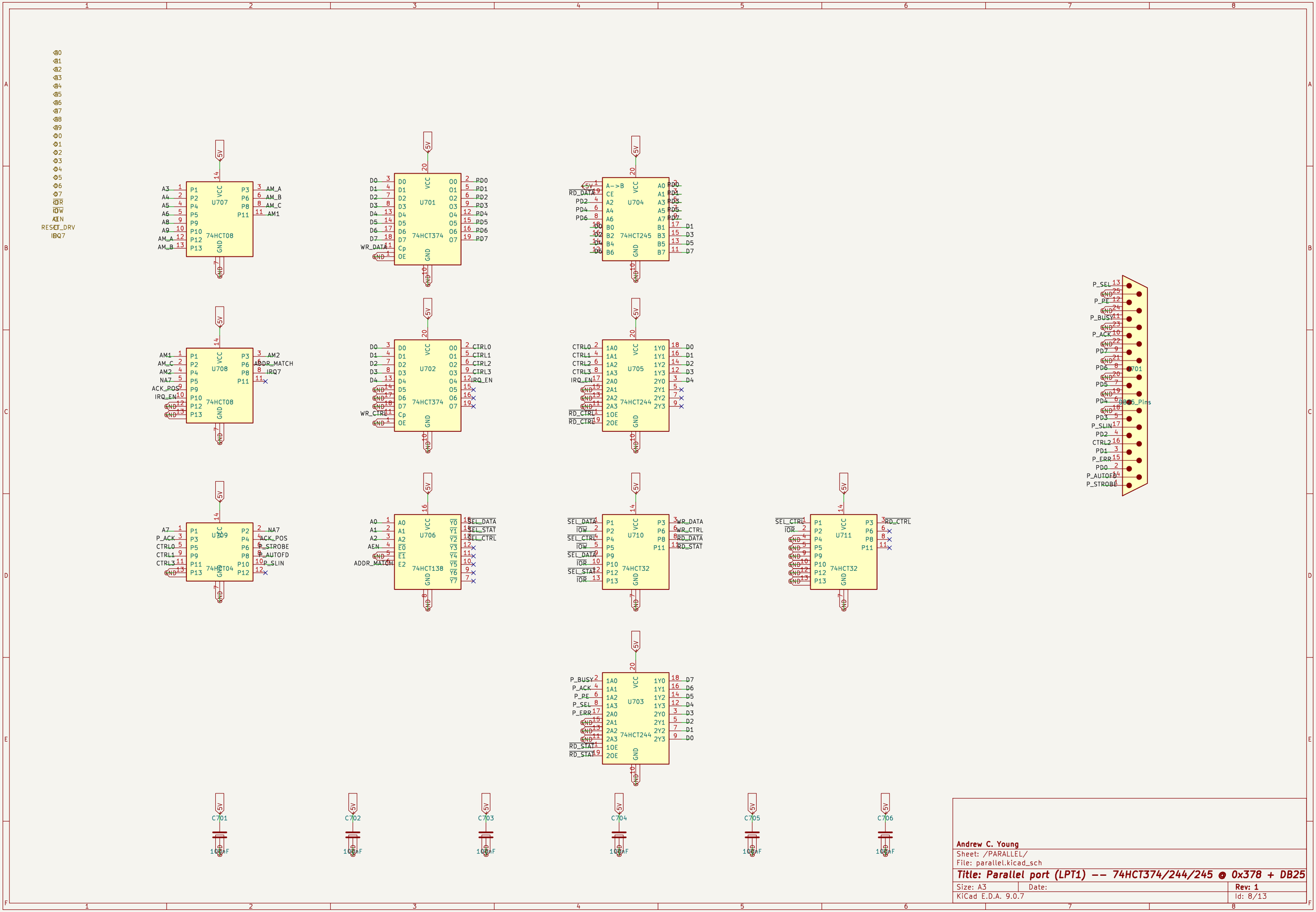
Andrew C. Young

Sheet: /COM2/

File: com_port.kicad_sch

Title: COM port -- 16C550 UART + MAX3241 + DB9 (instanced x2)

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 7/13



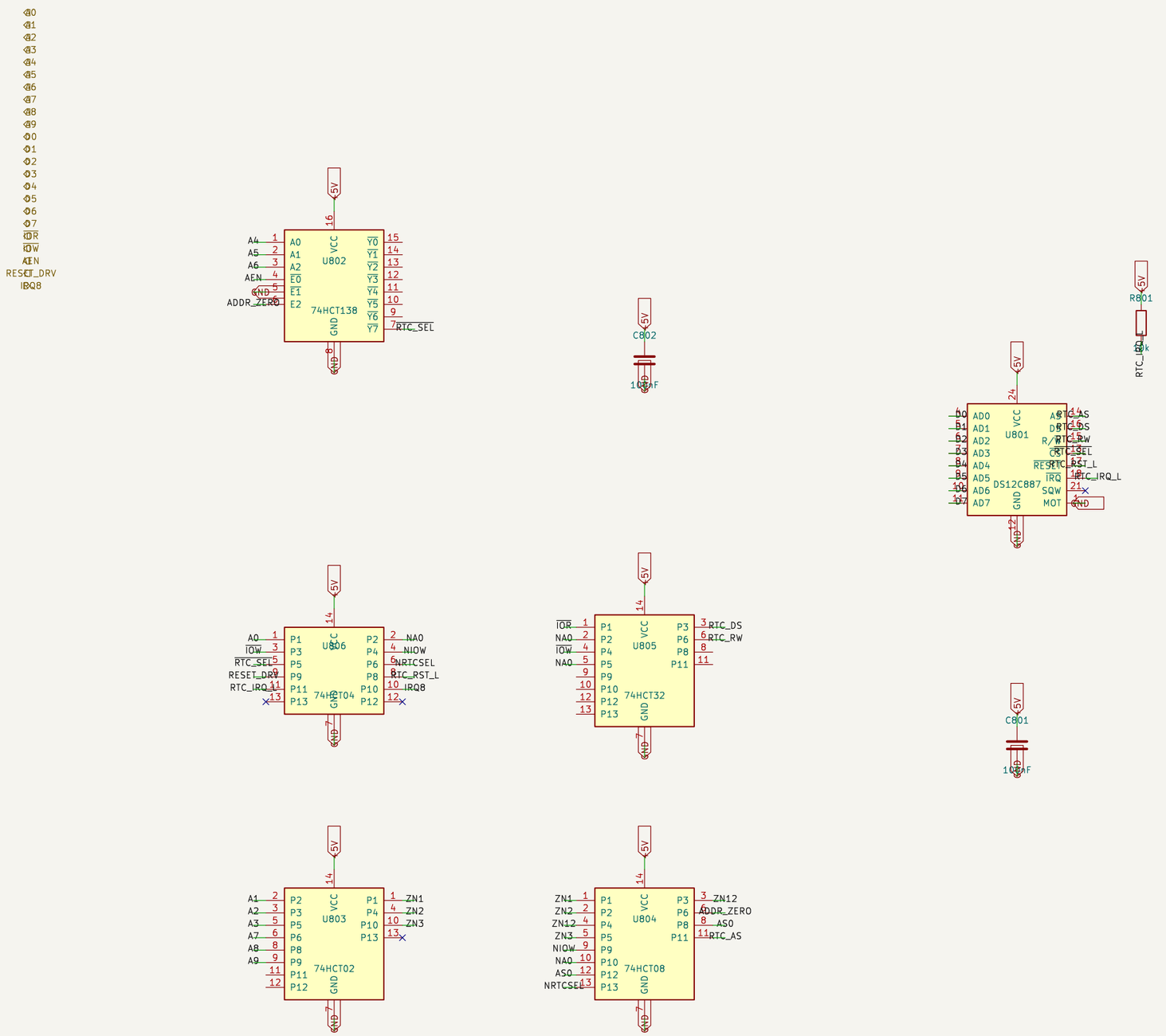
Andrew C. Young

Sheet: /PARALLEL/

File: parallel.kicad_sch

Title: Parallel port (LPT1) -- 74HCT374/244/245 @ 0x378 + DB25

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 8/13



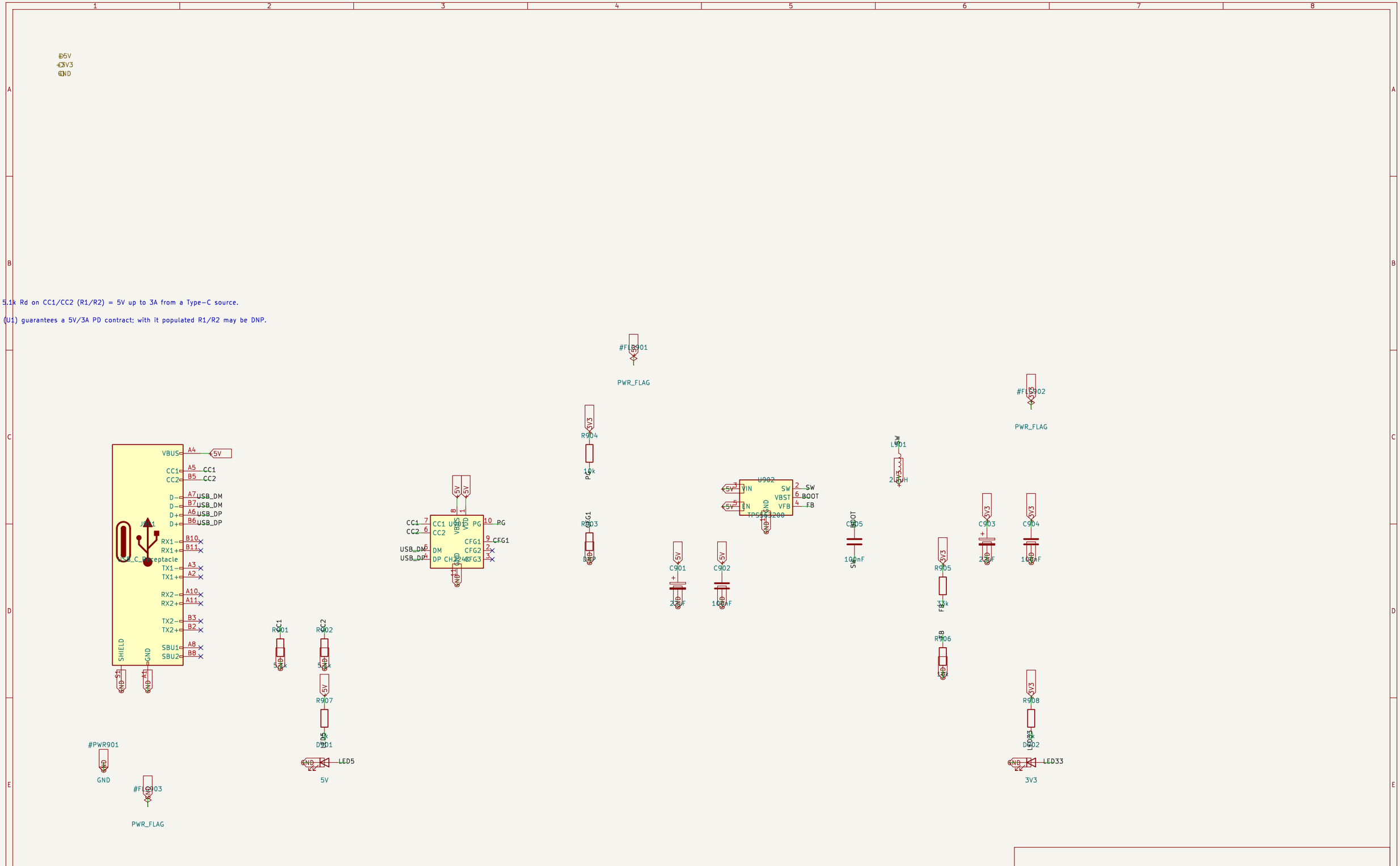
Andrew C. Young

Sheet: /RTC/

File: rtc.kicad_sch

Title: Real-time clock -- DS12C887 @ 0x70/0x71

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 9/13



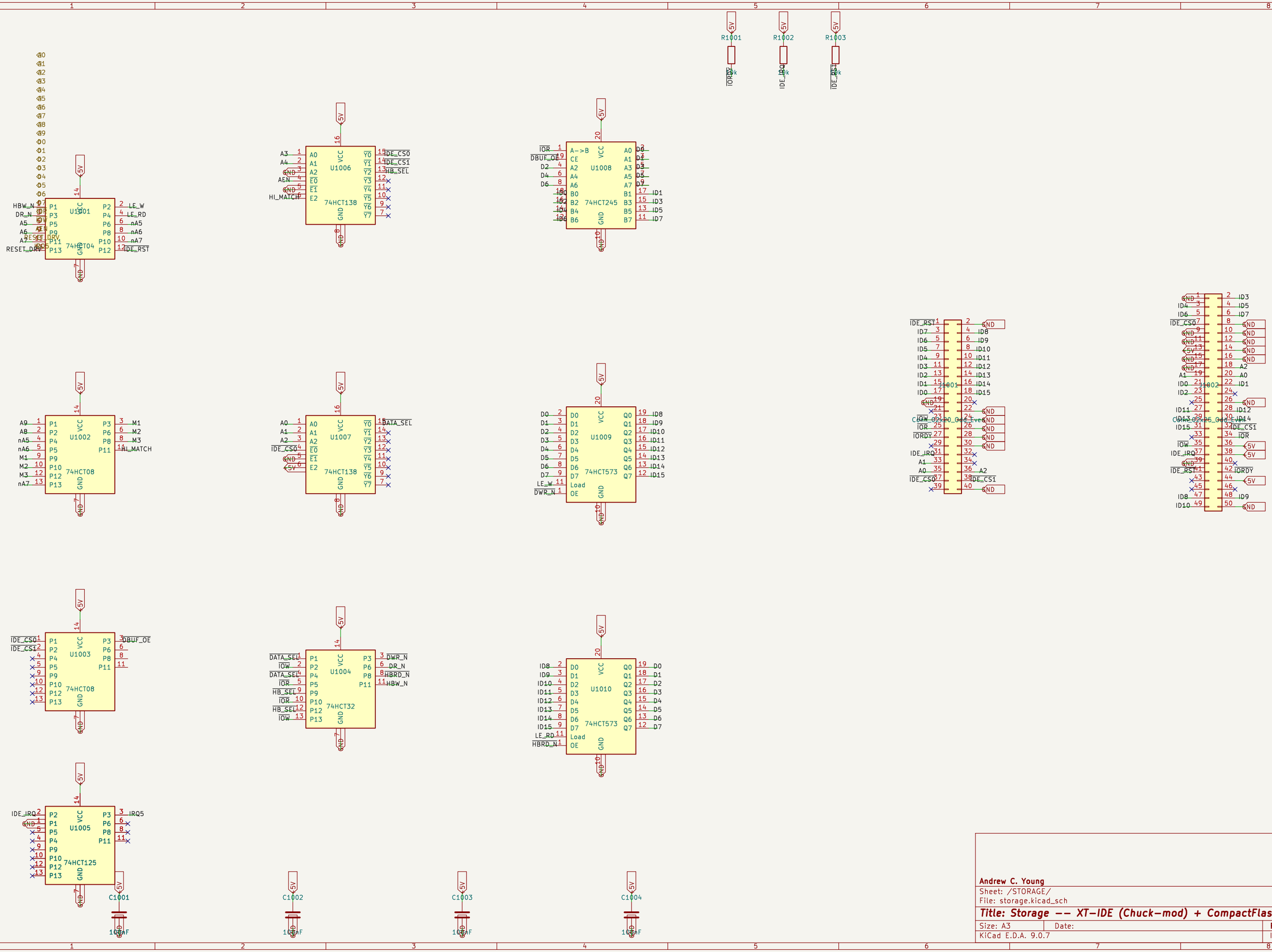
Andrew C. Young

Sheet: /POWER/

File: power.kicad_sch

Title: Power supply -- USB-C 5V in (CC/CH224K) -> 3.3V buck

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 10/13



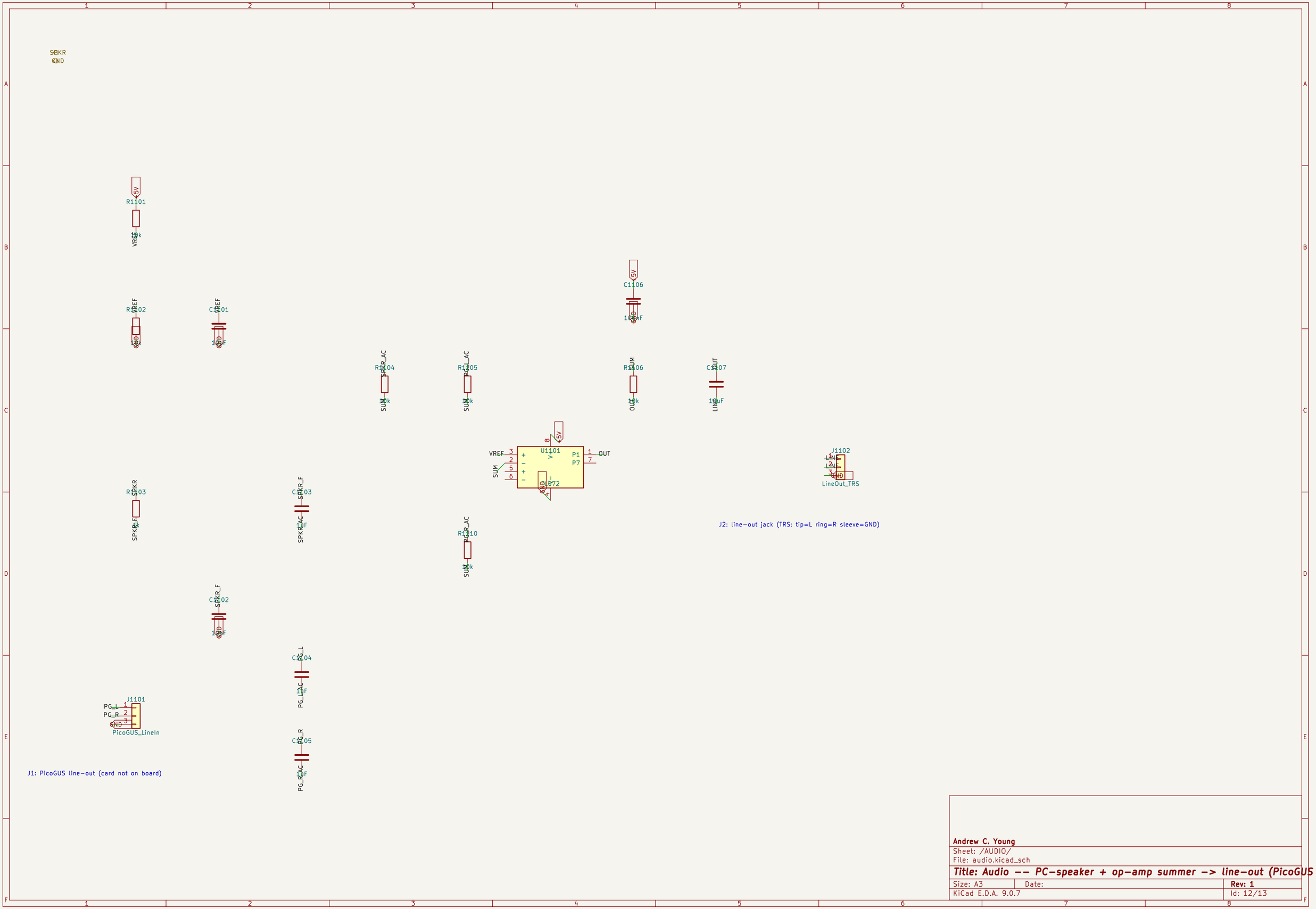
Andrew C. Young

Sheet: /STORAGE/

File: storage.kicad_sch

Title: Storage -- XT-IDE (Chuck-mod) + CompactFlash @ 0x300

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 11/13



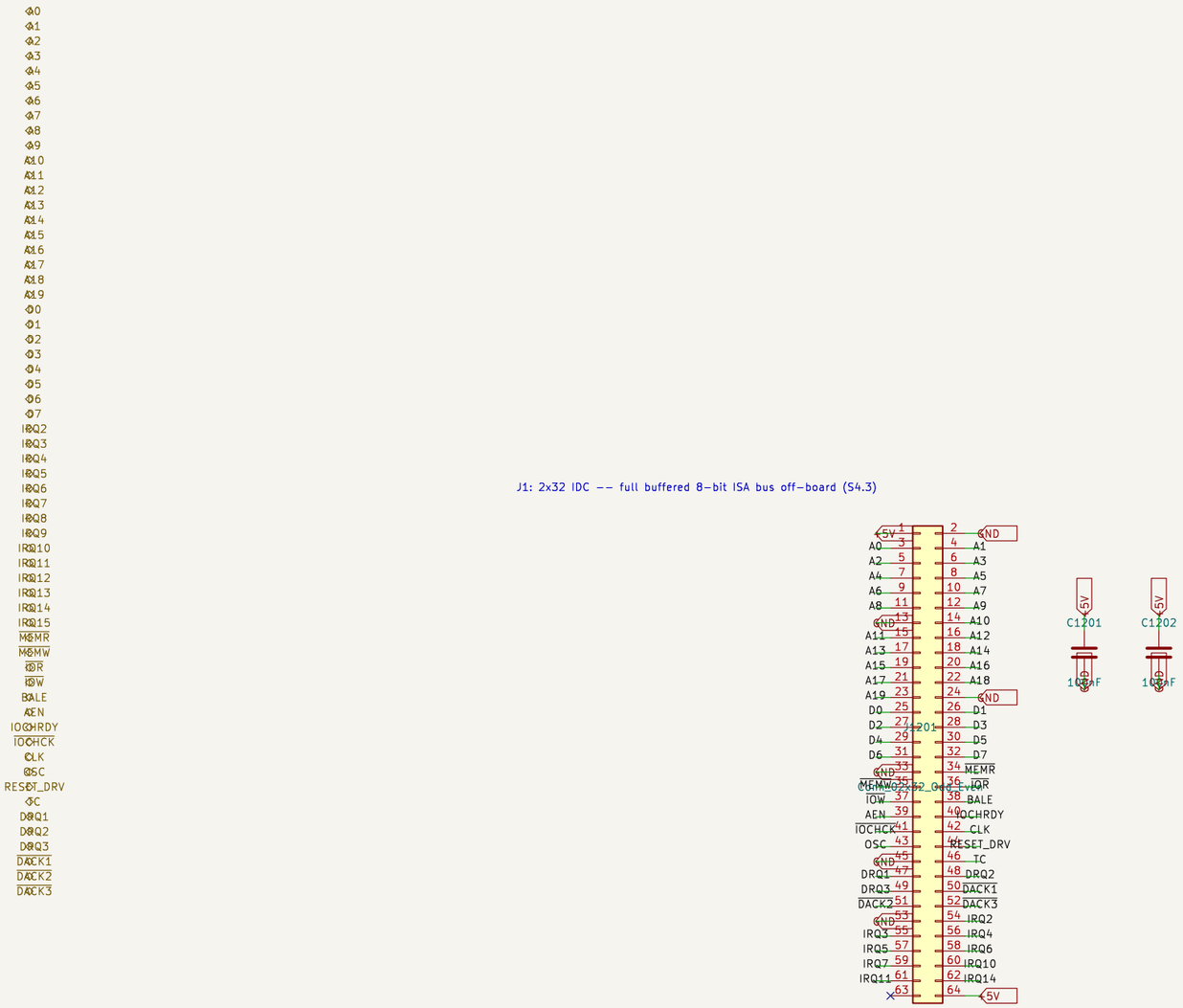
Andrew C. Young

Sheet: /AUDIO/

File: audio.kicad_sch

Title: Audio -- PC-speaker + op-amp summer -> line-out (PicoGUS line-

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 12/13



Andrew C. Young

Sheet: /SIDECAR/

File: sidecar.kicad_sch

Title: Sidecar -- 2x32 IDC ISA expansion header

Size: A3	Date:	Rev: 1
KiCad E.D.A. 9.0.7		Id: 13/13